

Changes to the ship workbench:



Loading a geometry

In addition to the four known ship types, you can load your own ship or geometry.

The following options are available for this purpose

- * Use sample ship

- * Import a geometry as:

STL file, the shape must be closed, i.e., including deck and transom. If there is no closed shape, the STL file cannot be converted into a solid.

IGES or STEP files are usually solids.

GF / GF1 files; these point specifications are combined into frames and must be converted manually into solids. It is recommended to create a closed mesh using lofts and then convert it into a solid in the part workbench

- * Own geometry; this must be a solid and can then be saved and edited as a ship instance.

You can choose a unit of measurement during import or automatic conversion of the measurements. FreeCAD calculates internally in mm.

The structure is placed so that the coordinate origin is at L/2. This is rather unusual in shipbuilding; normally, the coordinate origin is at the rear plumb line in Europe and at the front plumb line in the USA. Just a note, as this coordinate origin is rather unusual. This must be taken into account, especially for tank positions and weight positions.



Ships-area and Hydrostatic, as well as Create New tank and Tank capacity curve are unchanged from the original version.

The major difference to the previous version is the weight calculation. Until now, the data was stored exclusively in the ship instance and was somewhat difficult to influence or edit.



Now, a central load condition spreadsheet has been created in which you can fill the tanks you have created, listing all weights and cargo. There are three different groups:

Tanks, with the corresponding density of the liquid, percentage fill, and free liquid surfaces



Weights: here, primarily unchanging weights are assumed, such as: light ship weight, crew & effects, provisions, stores. The create new ship weight button is intended for these weights.



Once tanks and weights have been defined, a spreadsheet with the weights can be created using the create new loading condition button. The fill levels of the tanks can be entered as a percentage in this spreadsheet.



In addition, cargo can be imported. A packing list is best suited for this purpose. Clicking on the spreadsheet icon opens the import function for a packing list. On the start page of the import function, you will be asked for the packing list file. If there are several sheets, select the desired sheet. Under the Columns tab, assign the columns for name, description, weight, length, width, and height. Under the Settings tab, assign the units m, mm, kg, t, etc. Then, click on the Start Import button to start the import

Cuboids with specified length/width/height are formed, the labeling is engraved, and at the same time these weights and positions (assumed COG L/2, W/2, H/2) are entered into the load condition.

It is possible to move the cuboids to where you want them, and a stowage plan can be generated in this way.



Various columns in the load condition are still empty. The next step is the initial calculation of the given load condition. To do this, press the Calculate Load Case button, and the total weight and moments are then calculated.



With the Sink and Trim vessels button, equilibrium will be calculated with GM of load condition. The first time you call this up, a spreadsheet with different drafts and trims is saved, so that when you call it up again, the iterations to determine the ship's floating position are much faster.



The GZ curve shows the current stability and relates it to SOLAS requirements to determine whether the ship meets the stability criteria. Min. GM due to a leak calculation is not taken into account here. Output will be the actual loading condition including a graph showing GZ curve and area under GZ curve and will be saved as pdf file on the hard disk.

Crane section:

Crane



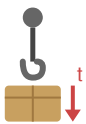
First of all, create a crane. A default crane will be created with its center at 0,0,0, all data like jib length or diameters can be edited in the crane data properties. When creating a crane one important part is the load capacity, you can choose an automatic load capacity curve with max outreach by given t and min outreach by given t or load stages (max3). Use placement to put crane base on the real position on board.



Couple crane to vessel, so crane will follow i.e. rolling angle of vessel



Decouple crane and vessel i.e. if you want to use another placing, you first have to decouple.



Single hook lift: Only one crane for lifting an item. Crane has to be chosen. If you have a second crane on board you can use this second crane as a sort of counter weight, you can choose this option in the second step. In the next step put in the weight to be lifted and if you want to have a calculation based on maximum outreach. Then click calculate. You will get max outreach and jib angle and another button will be active, to transfer this into the load condition for recalculation the stability, with the button Execute crane jib will show maximum position



Tandem lift will be similar to single hook lift, but of course for two cranes and load will be calculated for each crane. To have a feeling if this lift can be executed a small simulation can be initialized with the main focus to keep the distance of lifting points at the cargo. One point is necessary to put in the deck height above base to have the cargo free above the hatch. The radii will be shown for given crane load and you can choose how many steps you want to simulate. Every step can be put into the LoadCondition and will be calculated, so max outreach situation, maximum boom up situation and situation crossing midship line can be checked.



Will open a stability monitor so that you can see drafts and heeling angle during a crane operation. This monitor has its limit at 3° and 5° list. When above 3° monitor will get red, above 5° monitor will show n.a. degrees, a complete stability calculation has to be done.